

Bay — Feature Tracker

Pilot v1 (shipped) vs. Update v2 (backlog) · Melbourne + Brisbane pilot · Updated 2026-09-22

Pilot v1 — Shipped, live now

Finding parking

- GPS location detection with **continuous live tracking** (the map dot moves while driving, decoupled from the search anchor so it doesn't spam re-searches)
- Manual location search by country/state/city, worldwide city database
- Nearby parking search with adjustable radius (500m / 1km / 2km / 5km) and a visible dashed boundary showing exactly where search coverage ends
- Destination search — check parking near a shopping centre or address before heading out, not just current location
- Map view with colour-coded legal status pins (free/paid/restricted/accessible) and marker clustering so dense areas (100+ bays on one block) render as a readable count instead of an unreadable pile of pins
- List view as an alternative to the map
- Zone category filter (Paid/ticket, Time-limited, Permit holders only, Loading zone, Accessible) plus a "Free only" toggle
- Off-street car park capacity layer (opt-in) — 143 real commercial car parks, ~59,600 total spaces, distinct "P" markers, clearly labelled as capacity-only (no live availability published for these)

Accuracy of what's shown

- Real council open data, not guesses: Melbourne (on-street bay rules, Pay Stay + legacy zone signage) and Brisbane (sign locations + restriction hours)
- Timezone-aware rule evaluation — a spot's status is computed in *its own* local time, not the viewer's device time (matters for anyone checking a spot in a different timezone)
- Live in-ground sensor occupancy for ~5,700 Melbourne CBD bays — genuine hardware ground truth, synced automatically every 15 minutes entirely server-side (Postgres pg_cron + pg_net), independent of any device being open
- Sensor "stuck/offline" detection — distinguishes a sensor that's stopped reporting from a bay that's genuinely been occupied a long time, based on the sensor's own heartbeat rather than how long the reading has held
- Crowdsourced occupancy reporting (Mark occupied / Mark free) with GPS-proximity gating, per-user rate limiting, and corroboration-based confidence (a single report shows as "unconfirmed" until a second independent person agrees)
- Council bay reference numbers shown where the source data publishes them (~4,953 spots) — likely, not yet confirmed, the same number marked on the physical kerb
- Known, disclosed limits: coverage is City of Melbourne LGA + Brisbane CBD only (confirmed via exhaustive search — no neighbouring Melbourne council publishes comparable data); data is a periodic re-sync, not continuously live except the sensor layer; community reports carry an explicit "community-sourced, confirm with the physical sign" disclaimer

Reporting & community

- Report a new sign with a required photo, held in a moderation queue before going public
- Guide page — country-aware plain-language explainer for each sign type
- Privacy-preserving tester identity ("Test User 1, 2, 3...") — no IP address collection, by design

Feedback & growth infrastructure

- Quick in-app feedback form (My Activity tab)
- Standalone detailed feedback survey — usage-focused questions (where you started/were going, did Bay actually help, did info match reality, would you use it again) plus the single most useful question: "what made you stop using Bay?"
- One shareable landing page (Try Bay / Give feedback / coverage disclaimer / Play Store early-access waitlist) instead of separate links
- Lightweight usage tracking (a real "search happened" event, not just survey opinions) feeding a simple `pi lot_dashboard` SQL view: test users, actual searches, survey responses, waitlist signups, repeat users

Platform

- Android APK (debug, sideloadable) with real signing keystore configured for Play Store submission
- Web PWA, deployed and testable on any phone with no install
- Full source on GitHub (main branch = app, gh-pages branch = live web deploy)

Update v2 — Backlog, not built yet

Highest priority

- **"I've parked here" passive detection** — infer occupied/vacated from a user's own GPS dwelling at a spot, instead of a manual tap. The single highest-leverage fix for occupancy trust, since it doesn't depend on hitting a critical mass of users the way corroboration does
- Live availability for off-street car parks (currently capacity-only, no real-time free-space count — the council doesn't publish this, would need a different source or partnership)

Personalization

- Vehicle profile, saved places, office-hours-aware suggestions
- Push notifications for spot availability
- Time-travel filter — pick a future date/time and see what's legally free then, instead of only "right now"

Coverage expansion

- Neighbouring Melbourne councils, *if* any ever publish bay-level data (checked exhaustively as of Sep 2026 — none currently do)
- Other Australian cities (Sydney, Perth, Adelaide) — pending a real dataset search per city, not assumed
- International expansion
- iOS build

Trust & data quality

- In-app moderation/admin screen (currently handled via direct database access)
- Historical statistical occupancy baseline ("usually free around this time") once real usage volume exists — not meaningful yet at pilot scale
- Permit-zone map overlay for councils that only publish zone boundaries, not individual bays (different UI treatment than a point pin)

This is a living document — update it as features ship or the backlog changes, rather than treating it as fixed at launch.